



Minus Zero autonomous vehicle being tested on road

at any time. So, the same technology, which is highly data-dependent AI, that works elsewhere won't be a viable or safe solution for the traffic found in emerging markets, which becomes a major challenge.

Our vision is to build a more affordable and safer future of mobility for emerging markets with driverless taxis that can handle the tricky ODDs (operational design domains) found here.

What is the definition of Level 5 autonomy?

There are basically five levels of autonomy. Level 0 is where the driver is completely responsible for driving the car. Levels 1, 2, and 3 are driver-assistance systems, which means that if for any reason the driver loses focus that might cause him to lose control over the vehicle, the vehicle is capable of driving itself for a couple of seconds. This is where carmakers like Tesla are.

Level 4 autonomy is achieved when an autonomous vehicle can operate without any driver in geo-fenced areas or ODD, say urban cities with proper roads.

A vehicle with Level 5 autonomy can drive itself anywhere around the world, on-road or off-road, without the need for human intervention to supervise or take over in case of any anomaly. A car with Level 5 autonomy can handle any kind of traffic, whether it is a mountainous area or a jam-packed urban space like New Delhi's Sadar Bazaar. In fact, it can drive better than a human.

But a Tesla car can drive itself from one location to another and even follows traffic signals on the US roads. Wouldn't that be at least a Level 4?

No, because Tesla says that a driver must be there on the seat and keep his eyes on the road. In Level 4, the

driver is missing but the geographical locations are limited. So, Tesla can operate on Level 4, but it is not safe on that level. Even Tesla calls itself level 3 because the driver has to be there; they are still developing their technology.

In your video, where you demonstrated your tech on a 3-wheeler, nobody is sitting on the driver's seat in the 3-wheeler, but someone is sitting at the back with a laptop with wires connected to the 3-wheeler. So, was it really autonomous?

It was for safety and backup as we are still developing our tech. The biggest challenge when you're driving on a public road is that you have to ensure the safety of everyone around you. There could be chances of a camera or something failing, so the person sitting at the back had to be there as a precaution. The wires were connected to the braking mechanism so that we could disengage the autonomy system and apply the brake immediately in case of any anomaly. Fortunately, the driver did not have to intervene at any point and the prototype was able to handle itself properly.

How many trial runs have been done so far on public roads?

We have tested it numerous times in private testing areas like open grounds or private campuses. However, as of now, we could test it in limited capacity on a public road because, since there are no self-driving regulations in India right now, we are not legally allowed to do that.

What does the government need to do for startups like yours to provide you access to a testing environment?

It's a bit of a catch-22 situation right now. Since there are currently no

self-driving cars in the country, there are no regulations regarding the same. Hence, it's become challenging to develop and test autonomous vehicles on the road. This is a chicken-and-egg problem.

There are three aspects to the laws that need to be formulated, some of which have already been passed in the US states like Arizona, Phoenix, and California, and also in Singapore and Germany.

The first important aspect is how do they define the safety standards. Safety standards for standard vehicles include requiring airbags in the cars. Similar safety standards need to be set for self-driving cars as well.

Second comes the liability. For instance, if there is an accident involving the car, then who will be liable. Usually, in India, the owner of the vehicle is liable. But in self-driving cars, who will be liable: the passenger, the manufacturer, or the service provider?

And finally, we need laws on privacy, because self-driving cars often collect data while it is being driven. We can collect any amount of data during testing, but a privacy law pertaining to autonomous cars would determine what data we can collect. Some companies even collect the full video data while the car is being driven. But does that not infringe on the privacy of others? In a nutshell, the government needs to form regulations or guidelines for safety, liability, and privacy.

India is way behind in terms of passing such rules. If you take developed and developing countries, India is still in the last or second last position in terms of legal regulations, whereas countries where very few self-driving vehicle industries are established, like Singapore, have already passed rules for autonomous vehicles.

How do you see the landscape of autonomous vehicles? What's being done by other major OEMs?

Leading brands like Mercedes and others who have their vehicles in India are focusing on ADAS systems, like rear parking sensors, which fall under Level 1 autonomy. Nobody is targeting Level 4+ autonomy in India right now as per public sources, although there are a couple of startups and other corporates who want to explore this field. But the major challenge that even companies like Tesla face is the traffic in India, which is so dynamic and random that their systems will not be able to handle it. So, while such